

Quiz 2

Name(s):

In this quiz, you will practice proving security by reduction.

1. Let $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ be an encryption scheme that has indistinguishable encryptions against an eavesdropper. Prove that the following encryption scheme is also secure against an eavesdropper.

$$\text{Enc}'_k(m) = \overline{\text{Enc}_k(m)}$$

The following questions are meant to guide you through the proof. If you feel that you do not need them, you can just provide the full proof at the end.

- (a) How would you decrypt?
- (b) Write down the assumption you need to make to start the proof by reduction. What security game is $\mathcal{A}_c$ playing?
- (c) In order to prove security by a reduction, what is the adversary $\mathcal{A}_r$ that you need to construct, what game is it playing?
- (d) How would you construct $\mathcal{A}_r$ using $\mathcal{A}_c$?
- (e) Argue that $\mathcal{A}_r$ succeeds if $\mathcal{A}_c$ succeeds.